Year 12 Maths - Integration

Similar questions - Areas under and between curves

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The curve $y = 6x - x^2$ and the line $y = 2x$ intersect at the origin O and at the point A.

(a) Find the coordinates of A. **[3]**

(b) Find the area of the region enclosed between the curve and the line. **[4]**

(c) Find the area of the region enclosed between the curve and the x-axis, and hence the fraction of this area that lies above the line $y = 2x$. **[3]**

2 The curve $y = x^2 - 4x$ crosses the x-axis at O and at the point B. The region R_1 lies between the curve and the x-axis from O to B. The region R_2 lies between the curve, the x-axis and the line $x = 5$.

(a) Find the coordinates of B and sketch the curve, shading R_1 and R_2. **[3]**

(b) Find the area of R_1. **[3]**

(c) Find the area of R_2 and hence the total area of R_1 and R_2. **[3]**

(d) Explain why $\int_0^5 (x^2 - 4x)\,\mathrm{d}x$ does not give the total area, and state its value. **[2]**

Question 1

(a) $6x - x^2 = 2x$, so $x^2 - 4x = 0$, giving $x = 4$ (and $x = 0$ at O). $A = (4, 8)$

(b) $\int_0^4 (4x - x^2)\,\mathrm{d}x = \left[2x^2 - \frac{x^3}{3}\right]_0^4 = 32 - \frac{64}{3} = \frac{32}{3}$

(c) The curve meets the axis at $x = 0, 6$. $\int_0^6 (6x - x^2)\,\mathrm{d}x = \left[3x^2 - \frac{x^3}{3}\right]_0^6 = 108 - 72 = 36$. Fraction above the line $= \dfrac{32/3}{36} = \dfrac{8}{27}$

Question 2

(a) $x(x-4) = 0$, so $B = (4, 0)$. The curve is a U-shaped parabola with minimum at $(2, -4)$; R_1 is the region below the axis between 0 and 4, and R_2 is the small region above the axis between 4 and 5

(b) $\int_0^4 (x^2 - 4x)\,\mathrm{d}x = \left[\frac{x^3}{3} - 2x^2\right]_0^4 = \frac{64}{3} - 32 = -\frac{32}{3}$, so the area is $\frac{32}{3}$

(c) $\int_4^5 (x^2 - 4x)\,\mathrm{d}x = \left(\frac{125}{3} - 50\right) - \left(\frac{64}{3} - 32\right) = \frac{7}{3}$. Total area $= \frac{32}{3} + \frac{7}{3} = 13$

(d) Area below the axis counts as negative in an integral, so the two parts partly cancel: $\int_0^5 = -\frac{32}{3} + \frac{7}{3} = -\frac{25}{3}$